THADOMAL SHAHANI ENGINEERING COLLEGE
DEPARTMENT OF
Artificial Intelligence & Data Science

Name of Student : Neeraj Pandey

Class : SE

Division : S2

Roll Number : 73

Batch : S21

Subject : Fundamentals of Telecommunication Systems

Experiment Number : 01

Experiment Title : Study and Operation of a Basic Telephone System

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Experiment No. 01

Title: Study and Operation of a Basic Telephone System

Aim:

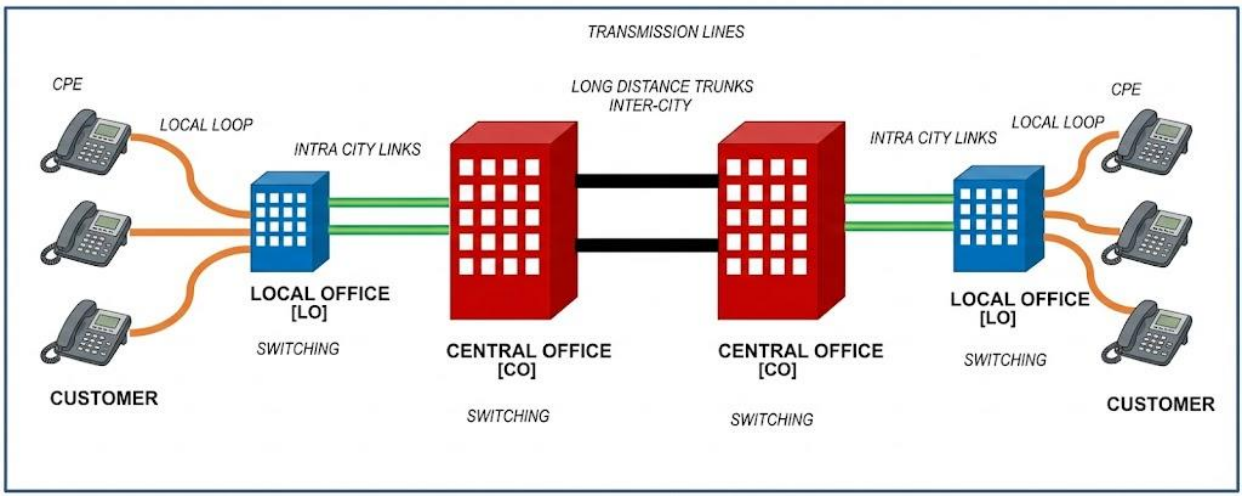
To study and understand the construction, components, and operation of a basic telephone system using online resources and videos.

Theory

A basic telephone system enables voice communication between two distant users by converting speech signals into electrical signals, transmitting them through a communication network, and reconverting them back into sound at the receiver.

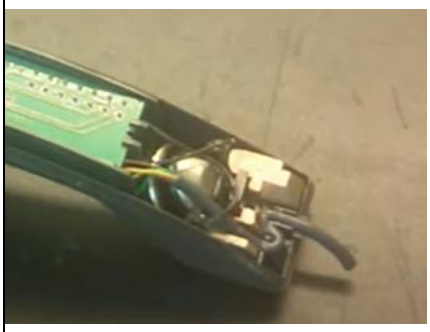
Traditional telephony operates on the Public Switched Telephone Network (PSTN), which uses circuit switching to establish a dedicated communication path for the duration of a call.

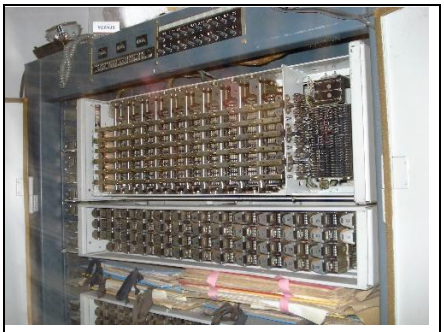
Block Diagram Instructions:



Task 1: The “Inside the Handset” Investigation

Table: Inside the Handset Components

Component	Visual Reference	Technical Function
Microphone		The microphone acts as an input transducer that converts sound waves produced by the human voice into corresponding electrical signals. It varies electrical resistance according to sound pressure, enabling voice transmission over the network.
Speaker		The speaker functions as an output transducer that converts incoming electrical signals into audible sound waves. It allows the listener to hear the reconstructed voice of the caller clearly.
Switch Hook		The switch hook detects whether the handset is placed on the cradle or lifted. When the handset is lifted, it completes the circuit with the exchange, indicating call initiation.
Ringer		The ringer alerts the user of an incoming call by producing an audible ringing sound. It is activated by an AC ringing voltage sent from the telephone exchange.

Component	Visual Reference	Technical Function
Keypad		The keypad allows the user to dial numbers by generating Dual Tone Multi Frequency (DTMF) signals. Each key produces a unique frequency pair recognized by the exchange.
Exchange		The exchange performs call switching, routing, and supervision. It connects calling and called subscribers by establishing a dedicated communication path.

Task 2: Basic Blocks of a Telephone System

Microphone (Transmitter):

The microphone converts acoustic speech signals into electrical signals. This conversion is necessary for transmitting voice information over long distances. It serves as the primary input device of the telephone system.

Local Loop:

The local loop is the physical copper wire connection between the subscriber's telephone and the telephone exchange. It carries voice signals, dialing signals, and ringing voltage. It forms the basic access network for subscribers.

Telephone Exchange:

The telephone exchange controls call setup, routing, supervision, and termination. It uses circuit switching to establish a dedicated path between subscribers. The exchange ensures efficient call management.

Transmission Network:

The transmission network connects multiple exchanges over long distances. It carries voice signals using optical fiber, coaxial cables, or microwave links. It ensures reliable and high-quality communication.

Receiver (Speaker):

The receiver converts incoming electrical signals back into sound waves. It enables the called or calling party to hear the transmitted voice clearly. It acts as the output device of the system.

Ringer and Dialing Circuit:

The ringer circuit alerts the user of incoming calls using an audible signal. The dialing circuit sends the dialed number to the exchange using electrical pulses or DTMF tones.

Task 3: Conceptual Working of a Telephone System

Off-Hook Condition:

When the handset is lifted, the switch hook closes the electrical circuit. This informs the exchange that the user intends to make a call.

Reception of Dial Tone:

After detecting the off-hook condition, the exchange sends a dial tone to the caller. This tone indicates that the system is ready to receive dialing information.

Dialing (DTMF Generation):

Pressing the keypad generates DTMF signals consisting of two frequencies. These tones are transmitted to the exchange to identify the dialed number.

Switching & Routing:

The exchange analyzes the dialed number and selects an appropriate route. It establishes a dedicated circuit between the calling and called subscribers.

Ringing State (Ring-back vs. Ringing):

The calling party hears a ring-back tone indicating that the call is progressing. The called party's telephone receives a ringing voltage that activates the ringer.

Voice Transmission (The Talking State):

Once the call is answered, both parties can speak and listen simultaneously. The microphone and speaker continuously convert voice signals between electrical and sound forms.

On-Hook (Call Termination):

When either user places the handset back on the cradle, the circuit is opened. The exchange detects this and releases the communication path.

Conclusion

A basic telephone system operates by converting physical voice signals into electrical signals using a microphone. These signals are transmitted through the local loop to the telephone exchange, where switching and routing occur across the PSTN network. At the receiving end, the electrical signals are reconverted into sound using a speaker. This experiment demonstrates how transducers, switching systems, and transmission networks work together to enable reliable voice communication.